

How to Select the Right Soldering/Desoldering Station



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Engineers know the value of solid and reliable soldering and desoldering stations. From carrying out simple soldering tasks to repairing and reworking surface-mount technology (SMT) boards, integrated circuits (ICs) and multi-layer boards, these stations are handy tools to have for every design engineer, technician, hobbyist or electronic enthusiast. In this article, we take a look at some of the latest features in soldering and desoldering stations, while highlighting the important parameters to consider before buying these for your applications.

Different kinds of stations available

Usage of soldering and desoldering stations range from very minute straight-forward jobs to heavy-duty applications. Based on the requirement and application, different stations are available, varying in power consumption, temperature range, microcontroller (MCU) based control and other features.

Contact soldering stations are the most common and popular kind of soldering sta-

tions. Abhishek Rao, technical marketing manager, element14 India, says, "These have a built-in power supply unit, which, in most cases, ensures the galvanic separation between the power circuit and the heating element." He adds, "Voltage on the heating element can be adjusted using the power supply unit and the heating temperature can be changed accordingly."

Lead-free soldering stations are another category that is used to wire lead-free printed circuit boards (PCBs). These usually make use of lead-free solders and are not necessarily expensive. With many developed nations making regulations favouring lead-free soldering, these stations could come in handy for future applications. Rao says, "Lead-free soldering stations are specially designed for working with lead-free solders, which offer good reliability and good solderability."

Although, lead-free stations have their own share of cons such as requirement of extra heat to melt lead-free solder, which, in turn, affects the tip of the soldering iron.

Most of the reputed manufacturers educate their buyers on ideal usage of lead-free soldering stations, which helps them maintain and extend the life of soldering tips.

Soldering stations are also packaged with different kinds of heaters. "Ceramic heaters, induction coil heaters and combination of heater, sensor and soldering tip integrated into one tool are some other variations in soldering stations," says V. Ramavallabhan, management and technical consultant, JVR Consultants.

Hot-air rework and desoldering stations are generally used for removing components from PCBs and re-soldering new ones. While rework stations are ideal for simple jobs, des-

Parameters to consider before buying a soldering station

Wattage and temperature control. The primary factor to consider while getting a soldering station is well-controlled soldering tip temperature. One should look for the temperature range one can set. Looking at the wattage of the soldering iron or pencil is also important. A soldering station with over 40W power would cover most applications.

Heat up time. For regular soldering, one wants the tip to reach the required temperature faster. To a certain extent, this is also related to the heat-recovery capability of the station. A station with slow thermal recovery may cause issues like cold solder joints.

ESD-safe design. Devices we deal with currently are more compact and usually more delicate than the ones that were used earlier. ESD damage accounts for a big portion of device failures during soldering. Sometimes even minor damages could lead to fail-leakage specifications. It is crucial to make sure that the unit has an ESD-safe design for long-term reliability.

Cost of soldering tips. It is advisable to be aware of soldering stations with expensive soldering tips as these are required to be changed often due to their daily wear and tear.

—Abhishek Rao, technical marketing manager, element14 India

